6

LAWS OF RETURNS

QUICK REVISION

1. Production: Production means the creation of goods or performance of services for sale in the market.
2. Production Function: The functional relationship between physical inputs and physical output of a firm is called production function.

or

Production function means functional relationship between inputs used and resulting output.

$$Qx = f(L, L \text{ and } K, E)$$

3. Types of Production Function: There are two types of production functions, vis.,
(i) Short-run Production Function: It refers to production in the short-run where some factors remain fixed and others variable. In short-run, product increases when more units of variable factors are used with the fixed factors. This is known as returns to a variable factors or law of variable proportions.

$$Qx = f(L, \sim K)$$

(ii) Long-run Production Function: It refers to production in a time period when all factors are variable. The effects of the changes in all the inputs on total output are called returns to scale or long-run production function.

$$Qx = (1, K \ldots)$$

4. Factors of Production: Various factors of production are classified into two categories: fixed factors and variable factors.
Fixed factors are those factors of production whose quantity remains fixed (or unchanged) at different levels of output, e.g., factory, building, machines, etc.
Variable factors refer to those factors of production which vary with every level of output, e.g., raw material ordinary labour power, fuel, etc.
In the short-run, only the quantity of variable factors can be changed while all the factors of production become variable in the long-run.

5. Concepts of Product: Generally, there are three main concepts of product which are used in context of input-output relations. These are:
(i) Total Product/Total Physical Product: It is defined as the total quantity of goods and services produced by a firm with the given inputs during a specified period of time. Total product is sum of total output of each unit of variable factor used in the process of production. Thus,

$$TP = \text{Sum of } MPs$$
$$TP = APXn$$

(ii) Marginal Product: It is a net addition to total product when one more unit of variable factors employed.

$$MP = TPn - TPn\ -1$$

$$MP = \Delta TP / \Delta L$$

(iii) **Average Product**: It is the per unit production of the variable factors, i.e.,

$$AP = TP/L$$

6.Law of Variable Proportions Of Returns to Factor: It states that as more and more units of a variable factor are applied to a given quantity of a fixed factor, the total product may increase at an increasing rate initially but eventually it will increase at a diminishing rate.

Assumptions:

(i) The law applies only in the short-run.
(ii) One factor of production is variable and others are fixed.
(iii) All units of variable factor are homogeneous.
(iv) State of technology is given and remains the same.
(v) Factor proportions can be changed.

There are 3 stages of Law of Variable Proportions:

(i) Increasing Returns: In this stage, TP increases at increasing rate and later at the diminishing rate. AP increases and reaches at its maximum.
MP initially increases then starts decreasing but continues to remain above AP

(ii) Diminishing Return: TP increases at a diminishing rate till it reaches at maximum point and then becomes constant. AP continues to fall. MP decreases and finally becomes zero (0).

(iii) Negative Returns: TP begins to fall, AP continues to fall but remains positive. MP becomes negative.

Causes of stages of law of variable proportions or Returns to a factor:

(i) Increasing Return to a Factor:
(a) Fuller utilisation of fixed factor
(b) Specialisation of labour

(ii) Diminishing Return to a Factor
(a) Imperfect factor substitutability
(b) Disturbing the optimum proportion

(iii) Negative Returns to a Factor:
(a) Overcrowding
(b) Management problems

1 MARK QUESTIONS

A. FILL IN THE BLANKS

1. Functional relationship between inputs and output is known as _____

2._____ and _____ are examples of fixed factors.

3. _____and_____ are examples of variable factors.

4. In _____ period all factors are variable.

5. In short period, law of _____ operates.

6. In long period, law of _____ operates.

7. Total product is_____ x units of variable factor.

8. Marginal product is _____

9. In law of variable proportion, the units of variable factor are assumed to be_____

10. When MP increases, TP increases at _____rate.

11. When MP falls, TP increases at _____

12. When MP becomes negative, TP _____

13. When marginal product is greater than average product, average product _____

14. When marginal product is equal to average product, average product is _____

15 _____is the slope of TP curve.

16. Two main causes of increasing returns to a factor are_____and _____

17. Both MP and AP are _____ shaped curve.

18. Two main causes of negative returns to a factor are _____ and _____

19. Shape of AP curve is _____

20. When a variable factor is combined with fixed factors, MP falls, it is known as _____

21. Fixed factors of production _____ during short run production.

22. Variable factor changes with _____ in short run production.

23. The point where TP starts increasing at diminishing rate is known as _____

ANSWERS

1. Production function	2. Land, Capital	3. Labour, Raw material	4. Long
5. Variable proportions, returns to a factor	6. Returns to a scale	7. AP	8. TP - TPn-1
9. Identical	10. Increasing	11. Diminishing rate	12. Falls
13. Rises	14. Constant	15. Inverted S	16. Optimum use of fixed factors, specialisation
17. Inverted U	18. Over crowding, Management problems	19. Inverted U	20. Diminishing returns
21. Remain constant	22. Change in output	23. Point of inflexion.	

B. TRUE OR FALSE

1. Total product will increase only when marginal product increases.
Ans. [False] Total product also increases when marginal product decreases but remains positive.

2. Increase in total product always indicates that there is increasing returns to a factor.
Ans. [False] Increase in total product diminishing returns to a factor.

3. When there are diminishing returns to a factor, marginal and total product both always fall.
Ans. [False] In case of diminishing returns to a factor, only marginal product falls and not total product.

4. When marginal product is zero total, product is at its maximum point.
Ans. [True] Zero marginal product signifies that change in total product is zero, it means total product has reached at its maximum.

5. In the operation of law of variable proportions, all the factors assumed to be variable.
Ans. [False] Law of variable proportions operates when one input is variable and all other inputs are fixed.

6. Shape of AP and MP curves is U.
Ans. [False] AP and MP curves are inverted U-shaped curves.

7. In the long run all factors of production are variable.
Ans. [True] In the long run, firm can adjust all its inputs to make changes in the output.

8. A rational producer aims to operate in first phase of law of variable proportions as total product increases at increasing rate.
Ans. [False] A rational producer aims to operate in second phase as total product is maximum and marginal product of each variable factor is positive.

9. When average product is maximum, marginal product will be greater than average product.
Ans. [False] When average maximum, marginal product will be equal to average product.

10. Under law of variable proportions factor ratio keeps on changing.
Ans. [True] The proportion of ratio between fixed and variable factor keeps on changing, when additional units of variable factors employed.

11. Production function does not depend upon the state of technology.
Ans. [False] Production function is always defined with respect to a given state of technology.

12. Marginal product always rises when average product is increasing.
Ans. [False] Marginal product can fall if average product is increasing as long as marginal product is more than average product.

13. Rising average product is possible only when MP is more than AP.
Ans. [True] When MP is more than AP, MP pulls AP up.

14. Marginal product cuts average product from its top.
Ans. [True] It happens because when AP rises, MP is more than AP. When AP falls, MP is less than AP so, MP cuts AP from its top.

15. When there are diminishing returns to a factor, total product always decreases.
Ans. [False] When there are diminishing returns to a factor, total product increases at a decreasing rate.

C. MULTIPLE CHOICE QUESTIONS

1. All the goods produced by a firm at a particular period of time are known as:
(a) total product (b) marginal product (c) average product (d) any of these

2. Basic unit of production in an economy is known as:
(a) consumer (b) producer (c) a unit (d) an economy

3. How total product is calculated?
(a) ∑AP (b) ∑TP (c) ∑MP (d) Any of these

4. Which of the following is the formula for calculating marginal product?
(a) TPn - TPn -1 (b) TP + TPn - 1 (c) TP - TPn + 1 (d) Any of these

5. With an increase in variable factors, whatever increase in production per unit of factor of production is known as:
(a) total output (b) marginal product (c) average product (d) any of these

6. How average product is calculated?
(a) LE (b) L/TP (c) SMP (d)ML/L

7. When total products falls, what is the tendency of marginal product?
(a) Falls (b) Negative (c) Positive (d) Zero

8. When total product increases at a diminishing rate, what will be the tendency of marginal product?
(a) Diminishes (b) Rises (c) Constant (d) none of these

9. When total product is maximum, marginal product would be:
(a) positive (b) zero (c) negative (d) None of these

10. Relationship between production and change in variable factors is known as:
(a) demand function (b) supply function (c) production function (d) none of these

11. Which of the following is a fixed factor of production?
(a) Labour (b) Raw material (c) Building (d) Electricity

12. Which of the following is a variable factor of production?
(a) Production manager (b) Machinery (c) Building (d) Labour

13. Average productivity curve (AP) is of which of following nature?
(a) Inverted U-shape (b) Inverted V-shape (c) U-shape (d) V-shape

14. Which of the following is a production function?
(a) Qx = Dx (b) Qx = f(A, B, C ....) (c) Qx = Px (d) None of these

15. When a variable factor is combined with fixed factors, MP falls. It is known as:
(a) law of increasing returns to a factor
(b) diminishing returns to a factor
(c) constant returns to a factor
(d) none of these

16. When a variable is combined with some fixed factors and MP increases at an increasing rate, it is known as:
(a) diminishing returns
(b) increasing returns
(c) constant returns
(d) any of these

17. Which of the following factors are varied during short period?
(a) Variable
(b) Fixed
(c) Both (a) and (b)
(d) None of these

18. Which of the following factors are varied during long period?
(a) Variable
(b) Fixed
(c) Both
(d) None of these

19. When average product can be zero or negative?
(a) Possible
(b) Sometimes possible
(c) Never possible
(d) None of these

20. Is marginal product can be zero or negative?
(a) Yes possible
(b) Not possible
(c) Sometimes possible
(d) All thrice conditions are possible

21. What is modern name of law of returns to factors?
(a) Law of equal proportions
(b) Law of diminishing proportions
(c) Law of increasing proportions
(d) Law of variable proportions

22. Law of variable proportions is applicable during which of following period?
(a) Short-run
(b) Long-run
(c) Medium period
(d) Any of these

23. Law of variable proportions is applicable during which of following period?
(a) Average and marginal product increase
(b) Average and marginal product fall
(c) Marginal product is zero
(d) Average product increases

24. Law of variable proportion explains three stages of production, at the end of second stage:
(a) average production rises and marginal product falls
(b) total product rises and marginal product is zero
(c) total product is maximum and marginal product is zero
(d) none of the above

25. What is the main cause of application of increasing returns?
(a) Specialisation
(b) Efficiency of factors
(c) Indivisibility of factors
(d) All of these

26. Main cause of law of diminishing marginal returns is:
(a) scarcity of factors
(b) imperfect substitution of factors
(c) both (a) and (b)
(d) none of these

27. When all the factors are increased in equal proportions but output increases at different proportions, it is known as:
(a) law of variable proportions
(b) law of returns to factors
(c) law of returns to scale
(d) any of these

28. From the given table MPP would be:

Variable Factor	TPP
1	40 Units
2	58 Units

(a) 40
(b) 58
(c) 18
(d) none of these

ANSWERS

1. (a)	2. (b)	3. (c)	4. (a)	5. (c)	6. (a)	7. (b)	8. (a)	9. (b)	10. (c)
11. (c)	12. (d)	13. (a)	14. (b)	15. (b)	16. (b)	17. (a)	18. (c)	19. (c)	20. (a)
21. (d)	22. (a)	23. (a)	24. (c)	25. (d)	26. (c)	27. (c)	28. (c)		

D. ASSERTION-REASON TYPE

Read the following statements-Assertion (A) and Reason (R), and select the correct alternative in each case:

(a) (A) is true, but (R) is false.
(b) (A) is false, but (R) is true.
(c) Both (A) and (R) are true and (R) is the correct explanation of (A).
(d) Both (A) and (R) are true but (R) is not the correct explanation of (A).

1. Assertion (A) : Variable factors can be changed in the short run.

Reason (R) : Variable factors are not required in case of zero output.

2. Assertion (A) : In law of variable proportion, one factor is variable and the other factor of production remains constant.

Reason (R) : This law operates in the short run.

3. Assertion (A) : TP first increases at increasing rate and then at a diminishing rate.

Reason (R) : MP first increases, reaches maximum and starts decreasing.

4. Assertion (A) : A rational producer always seeks to operate in phase II of law of variable proportions.

Reason (R) : In phase II of law of variable proportions, marginal product is negative.

5. Assertion (A) : The production function establishes a relation between input and output, which is economical in nature.

Reason (R) : The production function specifies either the maximum output that can be produced with the given inputs or the minimum quantity of inputs needed to produce a given level of output.

6. Assertion (A) : In the long run, production is done only by using the variable factors.

Reason (R) : All factors are variable in the long run.

ANSWERS

1. (a)	2. (c)	3. (c)	4. (d)	5. (c)

2 MARK QUESTIONS

1. What is production function?

Ans. Production function shows the maximum quantity of a commodity that can be produced per unit of time with given amount of inputs, when the best production technique available is used.

$$Qx = f(L, K)$$

2. What is marginal physical product?

Ans. Marginal physical product may be defined as the change in total product resulting from one additional unit of a variable product.

$$MPP = TPPn - TPPn\text{-}1$$

3. Define marginal product.

Ans. Marginal product is the same as marginal physical product, i.e., it is the addition to the total product resulting from one additional unit of a factor which is variable in nature.

4. What is meant by production?

Ans. Transforming input into output is known as production. Or Act of creating utility of goods and services is known as production.

5. Define the term total product.

Ans. **Total Product /Total Physical Product**: It is defined as the total quantity and services produced by a firm

with the given inputs during a specified period of time or total product is sum or total of output of each unit of variable factor used in the process of production. Thus

TP = Sum of MPs

TP = APXn

6. Define the term average product.

Ans. It is the per unit production of the variable factors, le.,

$$AP = \frac{YP}{L}$$

7. Define the term short-run.

Ans. A time period when some factor inputs are fixed while others are variable is termed as short-run. The production in short-run can be increased only by increasing quantity of variable factors.

8. Define the term long-run.

Ans. All factors of production become variable when the distinction between fixed and variable factor become irrelevant. The production in long-run can be increased by when factors of production.

9. What do you mean by fixed factors?

Ans. Fixed factors of production are those factors which inputs cannot be changed at different levels of output in the short period, e.g., land, machinery, etc.

10. What do you mean by variable factors of production?

Ans. Variable factors are those which can be changed to change the output even in the short run. It is directly related to the output. For example: Labour, raw material, etc.

11. What does the law of variable proportions state?

Ans. The law of variable proportions states that as more and more units of a variable factor are applied to a given quantity of a fixed factor, the total product may increase at an increasing rate initially, but eventually it will increase at a diminishing rate

12. Explain the types of production function.

Ans. **Short-run Production Function:** It refers to production in the short-run where some factors remain fixed and others variable. In short-run, product increases when more units of variable factors are used with the fixed factors. This is known as returns to a variable factors or law of variable proportions. Qx = f (L, K), K is capital which remains fixed in the short -run.

Long-run Production Function: It refers to production in a time period when all factors are variable. The effects of the changes in all the inputs on total output are called returns to scale or long-run production function.

Qx = f(L, K) both are variable in the long-run.

13. What is meant by indivisibility of a factor of production?

Ans. Indivisibility of a factor of production means that some factors are available only in some minimum size. Certain inputs particularly machinery, management, etc., are available in very big units and cannot be divided into small units even when production is less.

14. What is law of diminishing returns?
Ans. The law of diminishing returns states that with the increase in a variable factor, keeping all other factors constant, the marginal product of the variable factor diminishes after a certain level of production.

15. In which stage of the law of variable proportions marginal product and total product decline?
Ans. In the stage three of the law of variable proportions, the total product and marginal product both decline.

3-4 MARKS QUESTIONS

1. Mention two causes of diminishing returns to a factor of production.
Ans. The two causes of decreasing returns to a factor of production are:
(i) The optimum proportion in which the fixed and variable factors are combined for production may get disturbed when a larger quantity of variable factor is applied on the same quantity of fixed factors.
(ii) The production will face problems of lack of goods-supervision and overcrowding, causing diminishing returns.

2. What is meant by returns to a factor? What lead to increasing returns to a factor?
Ans. Returns to a factor relates to the behaviour of total output as one variable input, say labour is varied. It is a short-run concept.
Following factors lead to increasing returns to a factor:
(i) Optimum combination of factors: In the beginning when quantities of a variable factors are applied to fixed factors, the system moves towards achievement of optimum combination of factors because then underutilised fixed factor (building, machine, land, etc.) are better and more fully used leading to increasing returns.
(ii) Specialisation: A greater degree of division of labour greatly raises productivity of labour and other factors.

3. Give the assumptions of law of variable proportions.
Ans. (i) The law applies only in the short-run.
(ii) One factor of production is variable and others are fixed.
(iii) All units of variable factor are homogeneous.
(iv) State of technology is given and remains the same.
(v) Factor proportions can be changed.

4. In which stage should a producer operate? Why?
Ans. A rational producer would always prefer to operate in the 2nd stage. He will not stay in 1st stage because it is the stage of increasing returns and under no circumstances, he would like to operate under 3rd stage where the total product starts decreasing. The 1st and 3rd stages are called stages of economic absurdity hence a rational producer would like to operate in 2nd stage because it is the stage where AP and MP of variable factor are declining but remain positive. Thus, this is the best stage to work.

5. Give the reasons for diminishing return to a factor.
Ans. **(i) Imperfect Factor Substitutability**: Factors of production are imperfect substitutes of each other. More and more of labour, e.g., cannot be continuously used in place of additional capital. Accordingly diminishing returns to variable factor becomes inevitable.
(ii) Disturbing the Optimum Proportion: Continuous increase in application of variable factor along with

fixed factors beyond a point crosses the limit of ideal factor ratio. This results in poor coordination between the fixed and variable factors which causes diminishing return to a factor.

6. Give the reasons for negative return to a factor.

Ans. **Overcrowding:** When more and more variable factors are added to a given quantity of fixed factor, it will lead to over crowding and due to this MP of the labours decreases and it goes into negative.

Management Problems: When there are too many workers they may shift the responsibility to others and it becomes difficult for the management to coordinate with them. The labours avoid doing work. All these things lead to decrease in efficiency of laboures. Thus the output also decreases.

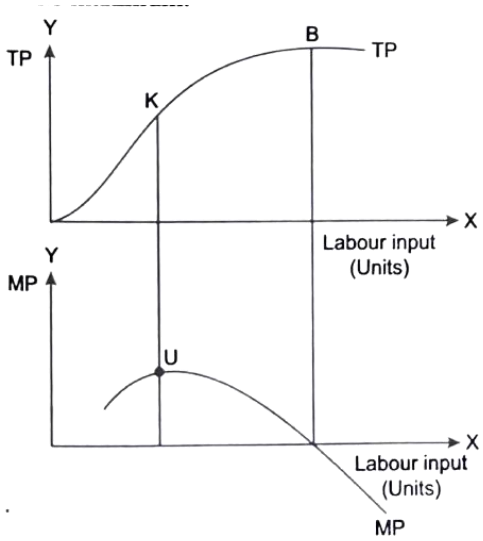
7. What are the factors which give rise to increasing returns to variable factors?

Ans. **(i) Fuller Utilization of the Fixed Factors:** Generally fixed factors are indivisible and underutilized. With greater application of variable factor these factors are better utilized its MPP tends to rise.

(ii) Increased Efficiency of Variable Factor: Application of specialisation and division of labour among the units of variable factors leads to greater efficiency and increase in MPP.

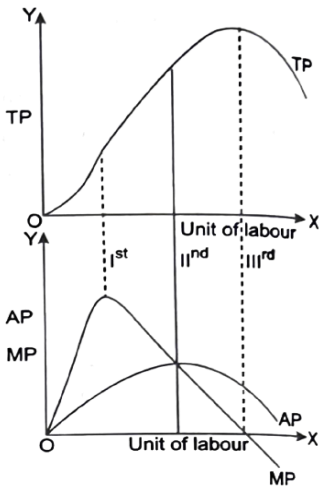
8. Draw a diagram to show the effect of total product on marginal product of a variable factor when total product reaches its maximum.

Ans.



6-8 MARKS QUESTIONS

1. Explain the three stages of law of variable proportions with the help of graph.



Ans. **Increasing Returns:** In this stage, TP increases at increasing rate and later at the diminishing rate.

AP increases and reaches at its maximum.

MP initially increases then starts decreasing but continues to remain above AP.

Diminishing Returns: TP increases at a diminishing rate till it reaches at maximum point and then becomes constant.

AP continues to fall.

MP decreases and finally becomes zero (0).

Negative Returns: TP begins to fall, AP continues to fall but remains positive.

MP becomes negative.

2. What do you understand by returns to factor? Why do diminishing returns to a factor operate?

Ans. Returns to a factor relates to the behaviour of total output as one variable input, say labour is varied. It is a short-run concept. There are three aspects of returns to a factor:

(i) Increasing returns to a factor,

(ii) Diminishing returns to a factor and

(iii) Negative returns to a factor.

Diminishing returns to a factor may occur due to following reasons:

(a) Fixity of the Factor: As more and more units of the variable factor continue to be combined with the fixed factor, the later gets over-utilized.

(b) Imperfect Substitution among Factor: Beyond a certain limit, factors of production cannot be substitute for one another, e.g., more and more of labour cannot be continuously used in place of additional capital.

3. Show the relationship between TP and MP.

Ans. Relationship between TP and MP (graph is given in Q.1 of this section):

(i) When TP increases at increasing rate, MP also increases.

(ii) When TP starts increasing at decreasing rate, MP decreases but remains positive.

(iii) When TP is maximum and constant, MP is 0 (zero).

(iv) When TP begins to fall, MP is negative.

4. Show the relationship between AP and MP.

Ans. Relationship between AP and MP (graph is given in Q.1 of this section)

(i) Both AP and MP curves are derived from TP since, $AP = \frac{TP}{L}$ and $MP = \frac{\Delta TP}{\Delta L}$

(ii) When MP is greater than AP, AP rises but MP rises at faster pace.

(iii) When MP equals to AP, AP is constant.

(iv) When MP is less than AP, AP falls but MP falls at higher rate.

5. Explain the law of variable proportions.

Ans. It states that as more and more units of a variable factor are applied to a given quantity of a fixed factor, the total product may increase at an increasing rate initially but eventually it will increase at a diminishing rate.

Assumptions:

(i) The law applies only in the short-run.

(ii) One factor of production is variable and others are fixed.

(iii) All units of variable factor are homogeneous.

(iv) State of technology is given and remains the same.

(v) Factor proportions can be changed. (graph is given in Q.1 of this section)

There are three stages of law of variable proportions:

(i) Increasing returns: In this stage, TP increases at increasing rate and later at the diminishing rate.

AP increases and reaches at its maximum.

MP initially increases then starts decreasing but continues to remain above AP.

(ii) Diminishing returns: TP increases at a diminishing rate till it reaches at maximum point and then becomes constant.

AP continues to fall.

MP decreases and finally becomes zero (0).

(iii) Negative returns: TP begins to fall, AP continues to fall but remains positive.

MP becomes negative.

QUESTIONS BASED ON DIFFERENTIATION

1. What is the difference between average product and marginal product?

Ans. Average product refers to the output per unit of variable factors. It is found by dividing the total product by the number of variable factors, e.g., number of labourers. Marginal product is the change in total product resulting from one additional unit of variable factor.

2. Differentiate between short-run production function and long-run production function.

Ans.

S. No.	Basis	Short-run Production Function	Long-run Production Function
1.	**Meaning**	It refers to a situation in which we study the behaviour of the output when more and more units of a variable factor are applied along with fixed quantities of other factors.	It refers to a situation in which we study the behaviour of the output when all the factors vary in the same proportion.
2.	**Time-period**	It relates to short period.	It is concerned with long period.
3.	**Factor proportion**	In this case, the proportion between fixed and variable factors goes on changing as more units of the latter are applied.	Factor proportion does not change at all.
4.	**Scale of Production**	Scale of production does not change here.	Scale of production changes.
5.	**The other name**	It is known as law of variable proportion or returns to a factor.	It is known as law of returns to a scale.
6.	**Formula**	Qx = f(L, K) L = Labour, it is variable, K = It remains fixed.	Qx =f(L,K) Both are variables.

3. Differentiate between short-run and long-run.

Ans.

S.No.	Basis	Short-run	Long-run
1.	**Meaning**	Short-run refers to a period in which output can be changed by changing only variable factors.	Long-run refers to a period in which output can be changed by changing all factors of production.
2.	**Classification**	Factors are classified as variable and fixed factor in the short-run.	All factors are variable in the long-run.
3.	**Price determination**	In the short-run, demand is more active in price determination as supply cannot be increased immediately with increase in demand.	In the long-run, both demand and supply play equal role in price determination as both can be increased.

4. Differentiate between fixed factors and variable factors.

Ans.

S.No.	Basis	Variable Factors	Fixed Factors
1.	**Meaning**	Those factors which can be changed in the short- run.	Those factors which cannot be changed in the short-run.
2.	**Relation with output**	They vary directly with output.	They do not vary directly with output.
3.	**Example**	Raw material, casual labour, poer, fuel, etc.	Building, plant and machinery, permanent staff, etc.

CASE BASED QUESTIONS

CASE 1

Production is known as the act of manufacturing goods and services. The resources which are used to produce goods and services are known as factors of production in economics. There are two time periods to do the production. The technical relationship between output and factors of production is known as production function. Production function is also of two types. One production function is known as

Answer the following questions based on the above paragraph:

1. How many factors of production are there? Name them.
2. Which are the two time periods of production?
3. production function is known as returns to a factor? What is the other name of it?
4.What are the three stages of the returns to a factor? Why the second stage is good for production?

Ans. 1. There are four factors of production : Land, Labour, Capital and Entrepreneur.

2. Short Run and Long Run.

3. Short run production function is known as returns to a factor. The other name of it is Law of variable factor of production.

4. The three stages of law of variable factor are : Increasing returns to a factor, Diminishing Returns to a factor, Negative returns to a factor.

The 1st and 3rd stages are called stages of economic absurdity hence a rational producer would like to operate in 2nd stage because it is the stage where AP and MP of variable factors are declining but remain positive and TP increases and becomes maximum in this stage only. Thus this is the best stage to work.

QUESTIONS WITH HIGH DIFFICULTY LEVEL

1. What happens to TP when MP is zero?

Ans. TP is maximum and constant.

2. What happens to MP when TP increases at decreasing rate?

Ans. MP falls but remains positive.

3. As the variable input is increased by one unit, total output falls. What would you say about marginal productivity of labour?

Ans. Marginal productivity of labour is negative

4. What happens to TP when marginal productivity of variable input is negative?
Ans. TP falls.

5. When is TP maximum in relation to MP?
Ans. When MP is zero.

6. What happens to MP/MPP when TPP is declining?
Ans. MPP declines and remains negative.

7. How does fall in MP affect TP?
Ans. TP increases at decreasing rate.

8. What does point of inflexion on TP curve indicate?
Ans. It indicates the point at which TP curve changes its shape from increasing rate to diminishing rate.

9. Can TP increase when MP is falling?
Ans. Yes, TP increases when MP is falling because MP is an addition to TP.

10. What is a general shape of AP and MP?
Ans. Inverted U.

11. What is the shape of MP during second stage of law of variable proportions?
Ans. Downward sloping.

12. At what point does the MP curve cuts the AP curve?
Ans. Where AP is highest and constant.

WORKSHEET FOR SELF EVALUATION

1. If TP of two workers is 100 units and of four workers is 220 units, what is the MP of the fourth unit? **(2 Marks)**

2. Show the point of inflexion on a TP curve. **(2 Marks)**

3. When AP falls, what is the relation between AP and MP? **(2 Marks)**

4. Can TP increases when MP is falling? **(2 Marks)**

5. Give reasons, state whether the following statements are true or false? **(3 Marks)**
(i) When there are diminishing returns to a factor, total production always decreases.
(ii) Total product will increase only when marginal product increases.
(iii) Total product and marginal product both decline in a situation of diminishing returns.

6. Identify the three phases in the law of variable proportions from the following and also give reason behind each phase. **(3 Marks)**

Units of Variable Input	1	2	3	4	5
Total Physical Product (TPP) Units	10	22	30	35	30

7. State the law of variable proportions. **(2 Marks)**

8. What is the reaction of AP when: (i) MP is more than AP; (ii) MP is less than AP; (iii) MP is equal to AP? **(2 Marks)**

9. On the basis of following table, find AP and MP. **(4 Marks)**

Labour	0	1	2	3	4	5
Total Product	0	15	35	50	40	48

10. What do you mean by imperfect substitutability? **(2 Marks)**